

3 Lecture 3

6. Elementary row matrix operations

Definition 3.1. Elementary row (matrix) operation of the first type, denoted $d_{r,\lambda}$, is a function which multiplies the r -th row of $A \in \text{Mat}(m, n, \mathbb{F})$ by a non-zero scalar $\lambda \in \mathbb{F}$. That is,

$$d_{r,\lambda}: A \mapsto d_{r,\lambda}(A), \quad \text{where} \quad [d_{r,\lambda}(A)]_{(i)} \stackrel{\text{def}}{=} \begin{cases} \lambda[A]_{(r)}, & \text{if } i = r; \\ [A]_{(i)}, & \text{if } i \neq r. \end{cases} \quad (3.1)$$

For example

$$d_{3,-2} \begin{bmatrix} -3 & 1 & 2 & -1 \\ 1 & 3 & 0 & 1 \\ 4 & 1 & 2 & 0 \end{bmatrix} = \begin{bmatrix} -3 & 1 & 2 & -1 \\ 1 & 3 & 0 & 1 \\ (-2) \cdot 4 & (-2) \cdot 1 & (-2) \cdot 2 & (-2) \cdot 0 \end{bmatrix} = \begin{bmatrix} -3 & 1 & 2 & -1 \\ 1 & 3 & 0 & 1 \\ -8 & -2 & -4 & 0 \end{bmatrix}.$$

Definition 3.2. Elementary row (matrix) operation of the second type, denoted $l_{r,s,\lambda}$, is a function which adds row s of $A \in \text{Mat}(m, n, \mathbb{F})$ multiplied by a scalar $\lambda \in \mathbb{F}$ to row r of A .

$$l_{r,s,\lambda}: A \mapsto l_{r,s,\lambda}(A), \quad \text{where} \quad [l_{r,s,\lambda}(A)]_{(i)} \stackrel{\text{def}}{=} \begin{cases} [A]_{(r)} + \lambda[A]_{(s)}, & \text{if } i = r; \\ [A]_{(i)}, & \text{if } i \neq r. \end{cases} \quad (3.2)$$

For example

$$l_{1,2,3} \begin{bmatrix} -3 & 1 & 2 & -1 \\ 1 & 3 & 0 & 1 \\ 4 & 1 & 2 & 0 \end{bmatrix} = \begin{bmatrix} -3 + 3 \cdot 1 & 1 + 3 \cdot 3 & 2 + 3 \cdot 0 & -1 + 3 \cdot 1 \\ 1 & 3 & 0 & 1 \\ 4 & 1 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 10 & 2 & 2 \\ 1 & 3 & 0 & 1 \\ 4 & 1 & 2 & 0 \end{bmatrix}.$$

Definition 3.3. Elementary row (matrix) operation of the third type, denoted $t_{r,s}$, is a function which interchanges rows r and s of $A \in \text{Mat}(m, n, \mathbb{F})$. That is,

$$t_{r,s}: A \mapsto t_{r,s}(A), \quad \text{where} \quad [t_{r,s}(A)]_{(i)} \stackrel{\text{def}}{=} \begin{cases} [A]_{(s)}, & \text{if } i = r; \\ [A]_{(r)}, & \text{if } i = s; \\ [A]_{(i)}, & \text{if } i \neq r \text{ and } i \neq s. \end{cases} \quad (3.3)$$

For example

$$t_{1,3} \begin{bmatrix} -3 & 1 & 2 & -1 \\ 1 & 3 & 0 & 1 \\ 4 & 1 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 1 & 2 & 0 \\ 1 & 3 & 0 & 1 \\ -3 & 1 & 2 & -1 \end{bmatrix}.$$

These three functions are called elementary row matrix operations; if the type is not important, we usually denote such a function by e .

The following statement shows that the elementary row operations of the third type can be expressed through the elementary row operations of the first and second types (it means that we introduced the functions $t_{r,s}$ for convenience, not out of necessity).

Statement 3.1. *For any $A \in \text{Mat}(m, n, \mathbb{F})$ we have*

$$t_{r,s}(A) = (d_{r,-1} \circ l_{r,s,-1} \circ l_{s,r,1} \circ l_{r,s,-1})(A). \quad (3.4)$$

Proof. For a matrix $A = [\dots, A_{(r)}, \dots, A_{(s)}, \dots] \in \text{Mat}(m, n, \mathbb{F})$ (in this notation we omitted all rows except rows r and s) we have

$$\begin{aligned} [\dots, A_{(r)}, \dots, A_{(s)}, \dots] &\xrightarrow{l_{r,s,-1}} [\dots, A_{(r)} - A_{(s)}, \dots, A_{(s)}, \dots] \\ &\xrightarrow{l_{s,r,1}} [\dots, A_{(r)} - A_{(s)}, \dots, A_{(r)}, \dots] \\ &\xrightarrow{l_{r,s,-1}} [\dots, -A_{(s)}, \dots, A_{(r)}, \dots] \\ &\xrightarrow{d_{r,-1}} [\dots, A_{(s)}, \dots, A_{(r)}, \dots]. \end{aligned}$$

■

The next lemma follows immediately from Definitions 3.1, 3.2 and 3.3; we leave the proof as an exercise.

Lemma 3.1. *For any $A \in \text{Mat}(m, n, \mathbb{F})$ we have:*

$$\begin{aligned} (d_{r,\lambda} \circ d_{r,\lambda^{-1}})(A) &= A = (d_{r,\lambda^{-1}} \circ d_{r,\lambda})(A); \\ (l_{r,s,\lambda} \circ l_{r,s,-\lambda})(A) &= A = (l_{r,s,-\lambda} \circ l_{r,s,\lambda})(A); \\ (t_{r,s} \circ t_{r,s})(A) &= A. \end{aligned}$$

That is, the functions $d_{r,\lambda}$, $l_{r,s,\lambda}$ and $t_{r,s}$ are invertible.

Definition 3.4. We define the matrices $D_m(r, \lambda)$, $L_m(r, s, \lambda)$, $T_m(r, s) \in \text{Mat}_m(\mathbb{F})$ as follows:

$$\begin{aligned} D_m(r, \lambda) &\stackrel{\text{def}}{=} d_{r,\lambda}(I_m); \\ L_m(r, s, \lambda) &\stackrel{\text{def}}{=} l_{r,s,\lambda}(I_m); \\ T_m(r, s) &\stackrel{\text{def}}{=} t_{r,s}(I_m). \end{aligned}$$

(recall that I_m is the identity matrix of order m).

That is:

$D_m(r, \lambda)$ is a diagonal matrix, with diagonal entries 1 everywhere except in the r -th position, where the entry is $\lambda \in \mathbb{F} \setminus \{0\}$;

$L_m(r, s, \lambda)$ is the identity matrix I_m but with an $\lambda \in \mathbb{F}$ in the (r, s) -th position;

$T_m(r, s)$ is obtained by swapping row r and row s of the identity matrix I_m .

For example:

$$D_3(2, \lambda) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad L_4(1, 3, \lambda) = \begin{bmatrix} 1 & 0 & \lambda & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad T_4(1, 4) = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}.$$

Now it ought to be clear that for any $i, j \in [m]$ we have

$$[D_m(r, \lambda)](i, j) = \begin{cases} \lambda, & \text{if } i = j = r; \\ 1, & \text{if } i = j \neq r; \\ 0, & \text{if } i \neq j; \end{cases} \quad (3.5)$$

$$[L_m(r, s, \lambda)](i, j) = \begin{cases} \lambda, & \text{if } i = r \text{ and } j = s; \\ 1, & \text{if } i = j; \\ 0, & \text{otherwise.} \end{cases} \quad (3.6)$$

Definition 3.5. Matrices $D_m(r, \lambda)$, $L_m(r, s, \lambda)$ and $T_m(r, s)$ are called elementary matrices. One may say that an elementary matrix is a matrix which differs from the identity matrix by one single elementary row operation.

Due to the following theorem, we say that the matrix $D_m(r, \lambda)$, $L_m(r, s, \lambda)$ and $T_m(r, s)$ is the corresponding elementary matrix for the elementary row operation $d_{r,\lambda}$, $l_{r,s,\lambda}$ and $t_{r,s}$, respectively.

Lemma 3.2. *Elementary row operation $d_{r,\lambda}$, $l_{r,s,\lambda}$ and $t_{r,s}$ can be realized by multiplying the matrix $A \in \text{Mat}(m, n, \mathbb{F})$ on the left by the matrix $D_m(r, \lambda)$, $L_m(r, s, \lambda)$ and $T_m(r, s)$, respectively. That is, for any $A \in \text{Mat}(m, n, \mathbb{F})$ we have:*

$$d_{r,\lambda}(A) = D_m(r, \lambda)A; \quad (3.7)$$

$$l_{r,s,\lambda}(A) = L_m(r, s, \lambda)A; \quad (3.8)$$

$$t_{r,s}(A) = T_m(r, s)A. \quad (3.9)$$

Proof. Let us prove Equality (3.7).

If $i = r$ we have:

$$\begin{aligned} [D_m(r, \lambda)A]_{(r)} &\stackrel{(1.5)}{=} [D_m(r, \lambda)A](r, r) \cdot [A]_{(r)} + \sum_{\substack{k \in [m] \\ k \neq r}} [D_m(r, \lambda)A](r, k) \cdot [A]_{(k)} \\ &\stackrel{(3.5)}{=} \lambda \cdot [A]_{(r)} + \sum_{\substack{k \in [m] \\ k \neq r}} 0 \cdot [A]_{(k)} \\ &= \lambda \cdot [A]_{(r)} \\ &\stackrel{(3.1)}{=} [d_{r,\lambda}(A)]_{(r)}. \end{aligned}$$

If $i \neq r$ we have:

$$\begin{aligned} [D_m(r, \lambda)A]_{(i)} &\stackrel{(1.5)}{=} [D_m(r, \lambda)A](i, i) \cdot [A]_{(i)} + \sum_{\substack{k \in [m] \\ k \neq i}} [D_m(r, \lambda)A](i, k) \cdot [A]_{(k)} \\ &\stackrel{(3.5)}{=} 1 \cdot [A]_{(i)} + \sum_{\substack{k \in [m] \\ k \neq r}} 0 \cdot [A]_{(k)} \\ &= [A]_{(i)} \\ &\stackrel{(3.1)}{=} [d_{r,\lambda}(A)]_{(i)}. \end{aligned}$$

Thus, Equality (3.7) is proven.

We leave the proof of Equality (3.8) as an exercise.

Now Equality (3.9) follows immediately from Equalities (3.7), (3.8) and (3.4):

$$\begin{aligned} t_{r,s}(A) &= (d_{r,-1} \circ l_{r,s,-1} \circ l_{s,r,1} \circ l_{r,s,-1})(A) \\ &= D_m(r, -1)L_m(r, s, -1)L_m(s, r, 1)L_m(r, s, -1)A \\ &= (D_m(r, -1)L_m(r, s, -1)L_m(s, r, 1)L_m(r, s, -1)I_m)A \\ &= (d_{r,-1} \circ l_{r,s,-1} \circ l_{s,r,1} \circ l_{r,s,-1})(I_m)A \\ &= t_{r,s}(I_m)A \\ &= T_m(r, s)A. \end{aligned}$$

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Lemma 3.3. *Any elementary matrix is invertible. More precisely:*

$$D_m(r, \lambda)^{-1} = D_m(r, \lambda^{-1}); \quad (3.10)$$

$$L_m(r, s, \lambda)^{-1} = L_m(r, s, -\lambda); \quad (3.11)$$

$$T_m(r, s)^{-1} = T_m(r, s). \quad (3.12)$$

We are going to apply elementary row operations to simplify the form of a given matrix $A \in \text{Mat}(m, n, \mathbb{F})$ by reducing it to a matrix A' having a form that is (in a sense) as close to the matrix $\text{diag}_{m,n}(1, 1, \dots, 1)$ as possible.

Reduced Row Echelon Form (RREF)

Definition 3.6. We say that $[A]_{(i)}$ is a non-zero row of $A \in \text{Mat}(m, n, \mathbb{F})$ if $a_{ij} \neq 0$ for some $j \in [n]$; we say that $[A]_{(i)}$ is a zero row of A if $a_{ij} = 0$ for all $j \in [n]$.

Definition 3.7. We say that $[A]^{(j)}$ is a non-zero column of $A \in \text{Mat}(m, n, \mathbb{F})$ if $a_{ij} \neq 0$ for some $i \in [m]$; we say that $[A]^{(j)}$ is a zero column of A if $a_{ij} = 0$ for all $i \in [m]$.

Definition 3.8. The leading entry (or pivot element) of a non-zero row $[A]_{(i)}$ of $A \in \text{Mat}(m, n, \mathbb{F})$ is the first (from the left) non-zero element of $[A]_{(i)}$; a zero row has no leading entry.

Definition 3.9. A matrix $A \in \text{Mat}(m, n, \mathbb{F})$ is said to be in reduced row echelon form (RREF) when it satisfies the following conditions:

- 1) the leading entry of each non-zero row of A is equal to $1 \in \mathbb{F}$;
- 2) every zero row of A occurs below every non-zero row of A ;
- 3) if $1, \dots, r$ are non-zero rows of A , and if the leading entry of non-zero row i occurs in column k_i , for $i \in [r]$, then $k_1 < \dots < k_r$ (that is, each leading entry is in a column to the right of the leading entry in the previous row);
- 4) each column of A which contains the leading entry of some row has all its other entries $0 \in \mathbb{F}$.

That is, reduced row echelon matrix has the form

$$\begin{matrix} & 1 & & k_1 & & k_2 & & \dots & & k_{r-1} & & k_r & & n \\ \begin{matrix} 1 \\ \\ r-1 \\ r \\ r+1 \\ \\ m \end{matrix} & \left[\begin{array}{ccccccccccccccccccccccccc} 0 & \dots & 0 & 1 & * & \dots & * & 0 & * & \dots & * & 0 & * & \dots & * & 0 & * & \dots & * & * & * \\ 0 & \dots & 0 & 0 & 0 & \dots & 0 & 1 & * & \dots & * & 0 & * & \dots & * & 0 & * & \dots & * & * & * \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & \dots & 0 & 0 & 0 & \dots & 0 & 0 & 0 & \dots & 0 & 1 & * & \dots & * & 0 & * & \dots & * & * & * \\ 0 & \dots & 0 & 0 & 0 & \dots & 0 & 0 & 0 & \dots & 0 & 0 & 0 & \dots & 0 & 1 & * & \dots & * & * & * \\ 0 & \dots & 0 & 0 & 0 & \dots & 0 & 0 & 0 & \dots & 0 & 0 & 0 & \dots & 0 & 0 & 0 & \dots & 0 & 0 & 0 \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & \dots & 0 & 0 & 0 & \dots & 0 & 0 & 0 & \dots & 0 & 0 & 0 & \dots & 0 & 0 & 0 & \dots & 0 & 0 & 0 \end{array} \right],
\end{matrix} \quad (3.13)$$

where the asterisks, $*$, can be any elements of $\mathbb{F}$.

For example, the following matrices are in reduced row echelon form (the asterisks, $*$, can be any elements of $\mathbb{F}$, and $p = \min(m, n)$):

$$\text{diag}_{m,n}(\underbrace{1, 1, \dots, 1}_{k \text{ times } 1}, \underbrace{0, 0, \dots, 0}_{p-k \text{ times } 0}), \quad \begin{bmatrix} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & * & 0 & * & * \\ 0 & 0 & 0 & 1 & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Definition 3.10. A matrix $A \in \text{Mat}(m, n, \mathbb{F})$ is said to be in row echelon form when it satisfies the first three conditions of Definition 3.9.

Definition 3.11. Suppose that $B \in \text{Mat}(m, n, \mathbb{F})$ is in reduced row echelon form and there are elementary row operations $e_1, e_2, \dots, e_k$ such that $B = (e_1 \circ e_2 \circ \dots \circ e_k)(A)$. Then we say that B is the reduced row echelon form of A ; we denote B as $\text{rref}(A)$.

The following theorem is pivotal to this course, it shows that for any $A \in \text{Mat}(m, n, \mathbb{F})$ there exists at least one $\text{rref}(A)$. Note that it can be shown that any $A \in \text{Mat}(m, n, \mathbb{F})$ has the unique $\text{rref}(A)$.

Theorem 3.1. For any matrix $A \in \text{Mat}(m, n, \mathbb{F})$ there exist elementary row operations of the first and second types, $e_1, e_2, \dots, e_k$, such that

$$\text{rref}(A) = (e_1 \circ e_2 \circ \dots \circ e_k)(A).$$

Proof. First we reduce A into its row echelon form.

The following algorithm (sometimes it is called Gaussian elimination) transforms a given m -by- n matrix A into a matrix in row echelon form.

Step 1. If A is a zero matrix then we are done (indeed, it ought to be clear that a zero matrix is in RREF).

If $A = [a_{ij}]$ is a non-zero matrix, let $[A]^{(k)}$ be the first (i.e., the leftmost) non-zero column of A ($[A]^{(k)}$ exists since A is a non-zero matrix).

Let a_{ik} be the first (i.e., the topmost) non-zero entry of $[A]^{(k)}$ (a_{ik} exists since $[A]^{(k)}$ is a non-zero column).

If $i = 1$ then skip this step and go to Step 2.

If $i \neq 1$, apply function $t_{1,i}$ to A (that is, interchange row 1 and row i of A ; we can use functions $t_{r,s}$ due to Formula (3.4)). Let M_1 be the resulting matrix.

It should be clear that $[M_1](1, k) = a_{ik}$ is the leading entry of row 1 of M_1 , and that M_1 has the following form:

$$M_1 = \left[\begin{array}{cccc|ccc} 0 & \dots & 0 & a_{ik} & * & \dots & * \\ 0 & \dots & 0 & * & & & \\ \vdots & \ddots & \vdots & \vdots & & * & \\ 0 & \dots & 0 & * & & & \end{array} \right].$$

Go to Step 2.

Step 2. Apply function $d_{1, a_{ik}^{-1}}$ to M_1 (that is, multiply row 1 of M_1 by a_{ik}^{-1} ; we can do it since $a_{ik} \neq 0$). It should be clear that the resulting matrix, denote M_2 , has the following form:

$$M_2 = \left[\begin{array}{cccc|ccc} 0 & \dots & 0 & 1 & * & \dots & * \\ 0 & \dots & 0 & * & & & \\ \vdots & \ddots & \vdots & \vdots & & * & \\ 0 & \dots & 0 & * & & & \end{array} \right].$$

If $m = 1$ then we are done.

If $m > 1$ then go to Step 3.

Step 3. Let $\lambda_i = [M_2](i, k)$, for $2 \leq i \leq m$. Sequentially apply functions¹ $l_{i,1,-\lambda_i}$, where $2 \leq i \leq m$, to M_2 (that is, for each i , $2 \leq i \leq m$, add $-\lambda_i$ times row 1 to row i). It should be clear that the resulting matrix, denote M_3 , has the following form:

$$M_3 = \left[\begin{array}{cccc|ccc} 0 & \dots & 0 & 1 & * & \dots & * \\ 0 & \dots & 0 & 0 & & & \\ \vdots & & \ddots & \vdots & & & \\ 0 & \dots & 0 & 0 & & & \end{array} \middle| \begin{array}{c} A_1 \end{array} \right],$$

where A_1 is an $(m-1)$ -by- $(n-k)$ matrix.

Repeat Steps 1-3 for the matrix A_1 .

Now it should be clear that these steps allow us to transform a given m -by- n matrix A into a matrix in row echelon form (that is, a matrix satisfying the first three conditions of Definition 3.9). Therefore, we may assume that $A = [a_{ij}]$ is in row echelon form.

Let $a_{1,k_1}, \dots, a_{r,k_r}$ be the leading entry of row 1, $\dots$, r , respectively (note that, by Definition 3.10, $a_{1,k_1} = \dots = a_{r,k_r} = 1$). By Definition 3.10, the k_r -th column of A has the following form:

$$[A]^{(k_r)} = \begin{bmatrix} a_{1,k_r} \\ \vdots \\ a_{r-1,k_r} \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

Now we sequentially apply functions¹ $l_{i,r,-a_{i,k_r}}$, where $1 \leq i \leq r-1$, to A (that is, for each i , $1 \leq i \leq r-1$, we add $-a_{i,k_r}$ times row r to row i).

Now it should be clear that the resulting matrix, denote A_r , is in row echelon form, and that the k_r -th column of A_r has the following form:

$$[A_r]^{(k_r)} = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \begin{matrix} 1 \\ \\ r \\ \\ m \end{matrix}$$

It means that Condition 4) of Definition 3.9 is satisfied for the k_r -th column of A_r .

¹The exact order in which these functions are applied does not matter.

It remains to apply the analogous transformations to the columns $k_2, \dots, k_{r-1}$ of A_r . ■

In the following example, we show the main steps of the above proof of Theorem 3.1.
Let

$$A = \begin{bmatrix} 0 & 0 & 1 & 1 & -1 \\ 2 & 2 & 4 & 0 & 2 \\ -1 & -1 & 1 & 3 & -4 \end{bmatrix}.$$

First we transform A into a matrix in row echelon form.

Step 1. Since $k = 1$ and $i = 2$, we have

$$\begin{bmatrix} 0 & 0 & 1 & 1 & -1 \\ 2 & 2 & 4 & 0 & 2 \\ -1 & -1 & 1 & 3 & -4 \end{bmatrix} \xrightarrow{t_{1,2}} \begin{bmatrix} 2 & 2 & 4 & 0 & 2 \\ 0 & 0 & 1 & 1 & -1 \\ -1 & -1 & 1 & 3 & -4 \end{bmatrix} = M_1.$$

Step 2. Since $[M_1](1, 1) = a_{21} = 2$, we have

$$\begin{bmatrix} 2 & 2 & 4 & 0 & 2 \\ 0 & 0 & 1 & 1 & -1 \\ -1 & -1 & 1 & 3 & -4 \end{bmatrix} \xrightarrow{d_{1,2^{-1}}} \begin{bmatrix} 1 & 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 & -1 \\ -1 & -1 & 1 & 3 & -4 \end{bmatrix} = M_2.$$

Step 3. We have

$$\begin{bmatrix} 1 & 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 & -1 \\ -1 & -1 & 1 & 3 & -4 \end{bmatrix} \xrightarrow{l_{2,1,0}} \begin{bmatrix} 1 & 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 & -1 \\ -1 & -1 & 1 & 3 & -4 \end{bmatrix} \\ \xrightarrow{l_{3,1,1}} \left[\begin{array}{c|ccccc} 1 & 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 & -1 \\ 0 & 0 & 3 & 3 & -3 \end{array} \right] = M_3,$$

$$\text{where } A_1 = \begin{bmatrix} 0 & 1 & 1 & -1 \\ 0 & 3 & 3 & -3 \end{bmatrix}.$$

Now we are going to repeat Steps 1-3 for A_1 .

Step 1. Since $k = 2$ and $i = 1$, we skip this step.

Step 2. Since $[M_1](1, 2) = a_{12} = 1$, the function $d_{1,1}$ leaves the matrix unchanged:

$$\begin{bmatrix} 0 & 1 & 1 & -1 \\ 0 & 3 & 3 & -3 \end{bmatrix} \xrightarrow{d_{1,1}} \begin{bmatrix} 0 & 1 & 1 & -1 \\ 0 & 3 & 3 & -3 \end{bmatrix} = M_2.$$

Step 3. We have

$$\begin{bmatrix} 0 & 1 & 1 & -1 \\ 0 & 3 & 3 & -3 \end{bmatrix} \xrightarrow{l_{2,1,-3}} \left[\begin{array}{c|ccc} 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right],$$

where $A_1 = [0 \ 0 \ 0]$. Since A_1 has only one column (i.e. $m = 1$) we are done.

Note that the resulting matrix

$$\begin{bmatrix} \boxed{1} & 1 & 2 & 0 & 1 \\ 0 & 0 & \boxed{1} & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

is in row echelon form (we put the leading entries in boxes). Note that this matrix is not in RREF (indeed, its third column contains the leading entry and does not have all its other entries 0). We can easily amend it:

$$\begin{bmatrix} 1 & 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{l_{1,2,-2}} \begin{bmatrix} \boxed{1} & 1 & 0 & -2 & 3 \\ 0 & 0 & \boxed{1} & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

It remains to note that the last matrix is in RREF.

The next theorem follows immediately from Theorem 3.1 and Lemmas 3.2 and 3.3.

Theorem 3.2. *For any $A \in \text{Mat}(m, n, \mathbb{F})$ there exist elementary matrices of the types $D_m(r, \lambda)$ and $L_m(r, s, \lambda)$, E_1 , E_2 , $\dots$, E_k , such that*

$$A = E_1 \cdot E_2 \cdots E_k \cdot \text{rref}(A). \quad (3.14)$$